AIRSloths

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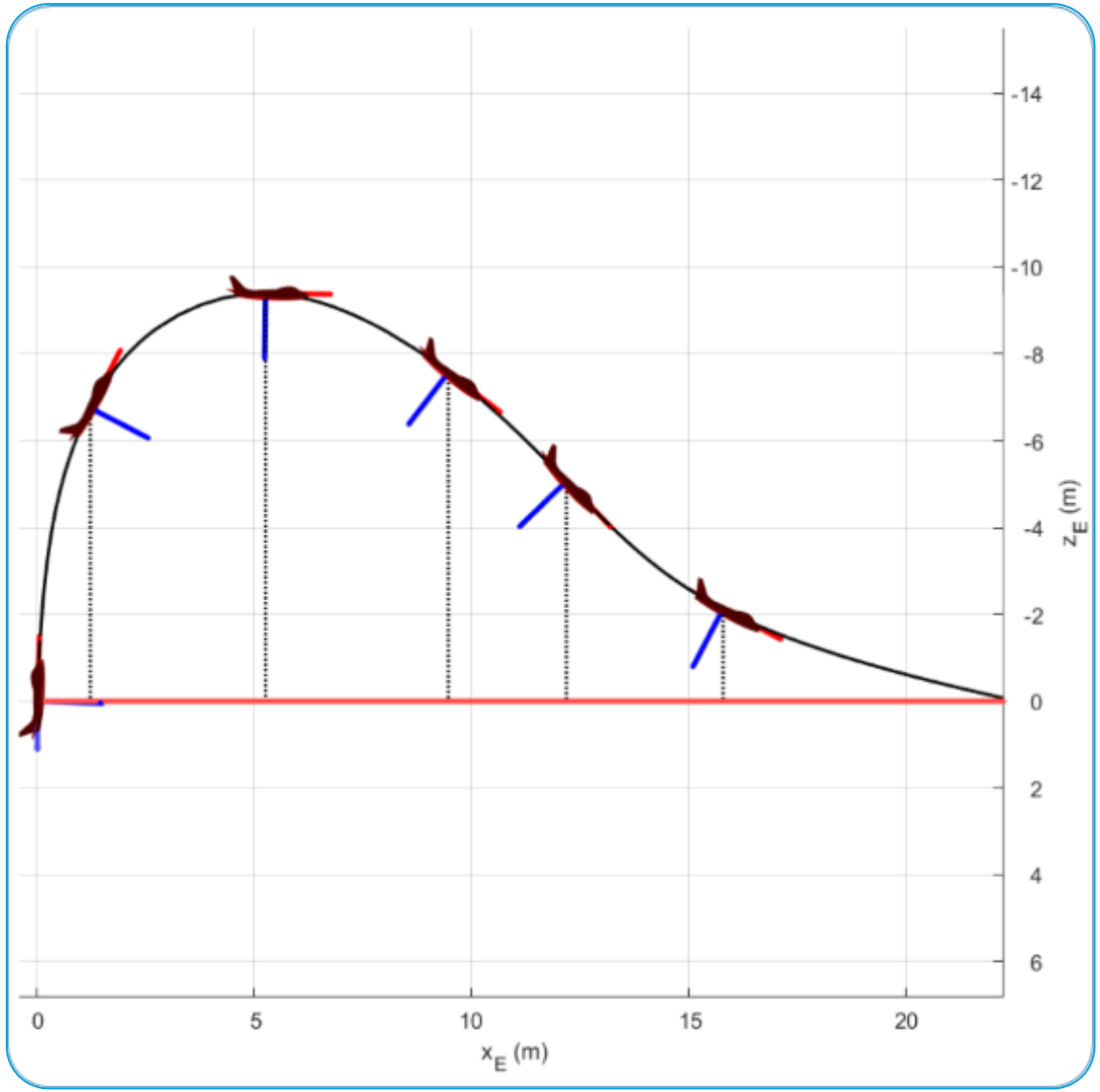
Contents

- Design
  - Goals
  - Aircraft
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- Sloshing
- Construction
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  - Safety
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- Lesson learnt

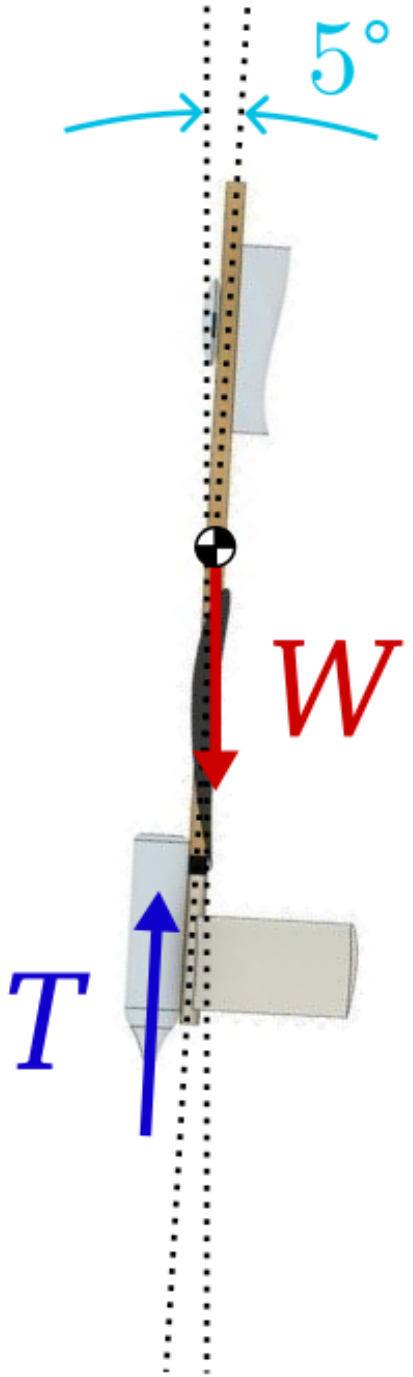
Design

Expected trajectory

- Ascent phase ($\mathcal{H}$)
- Slide phase
- Gliding phase ($\mathcal{R}$)

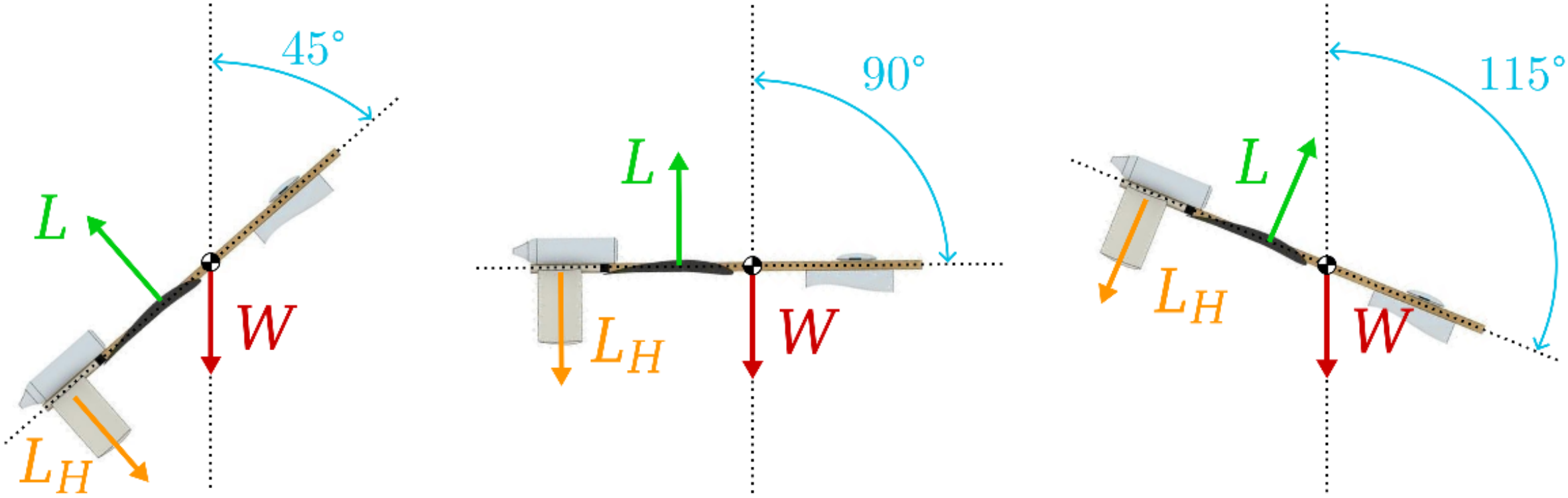






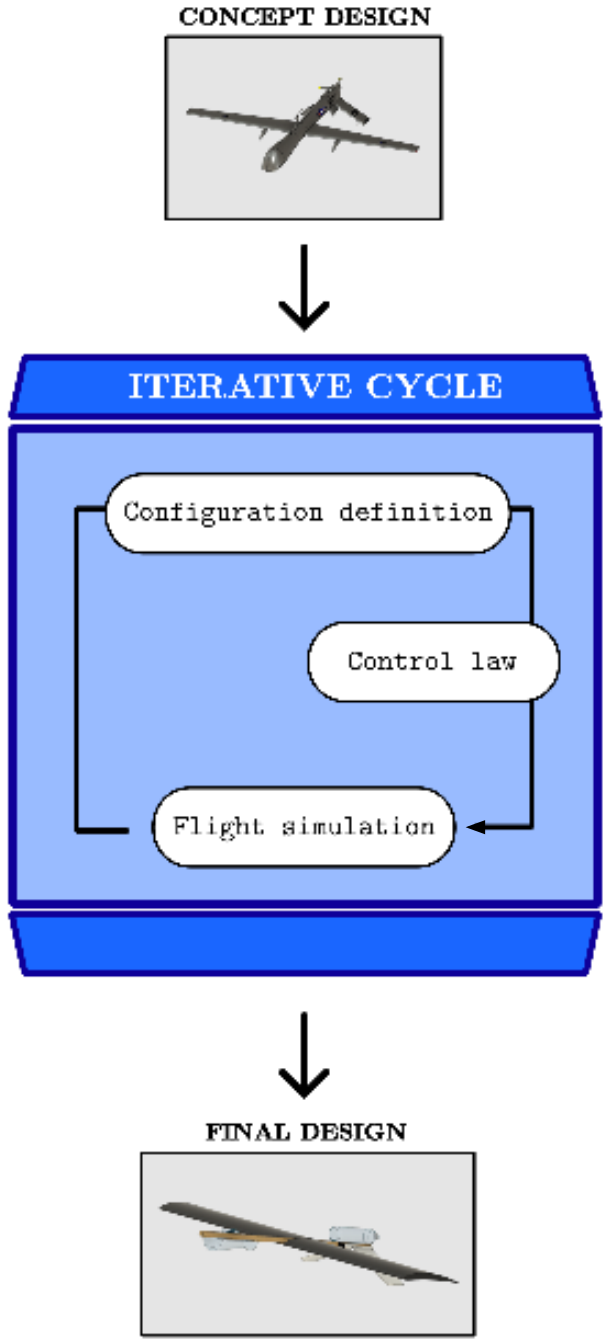
Physics involved

Intermediate phases



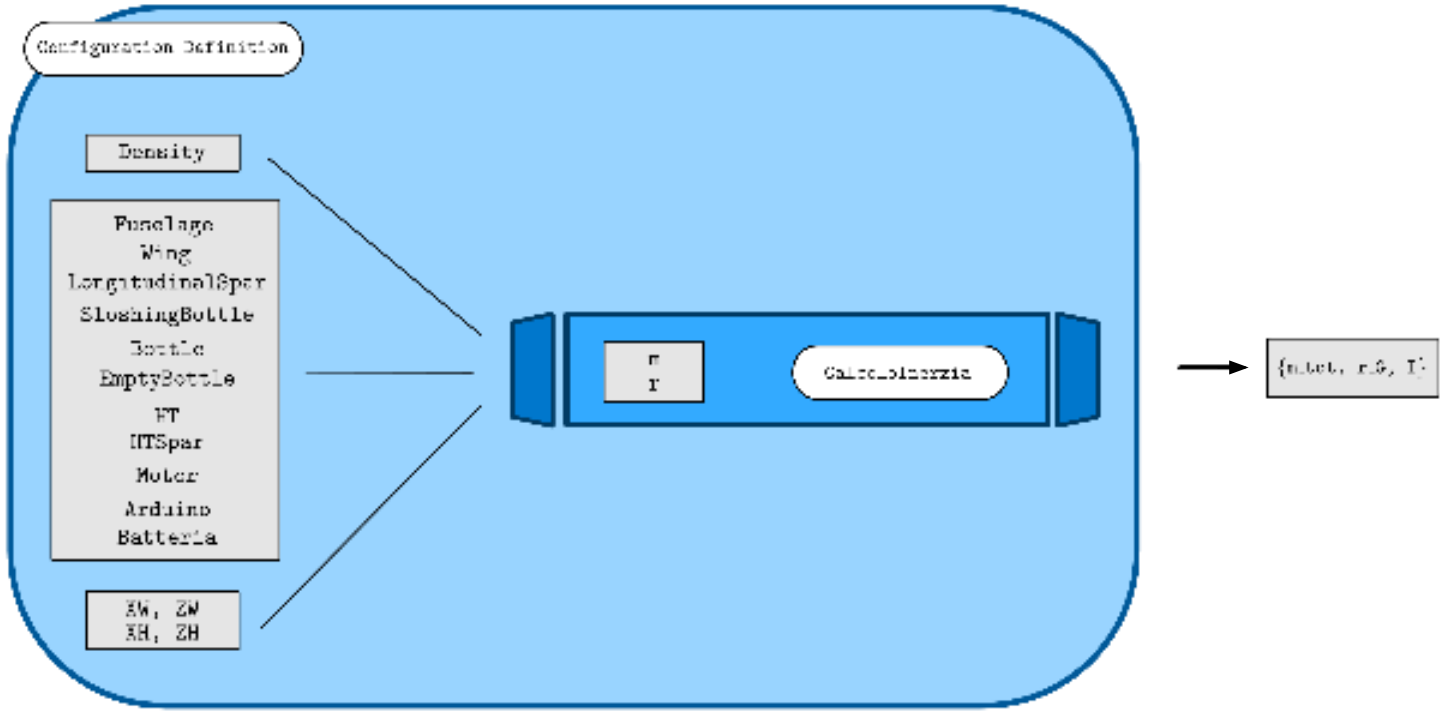
Flow chart

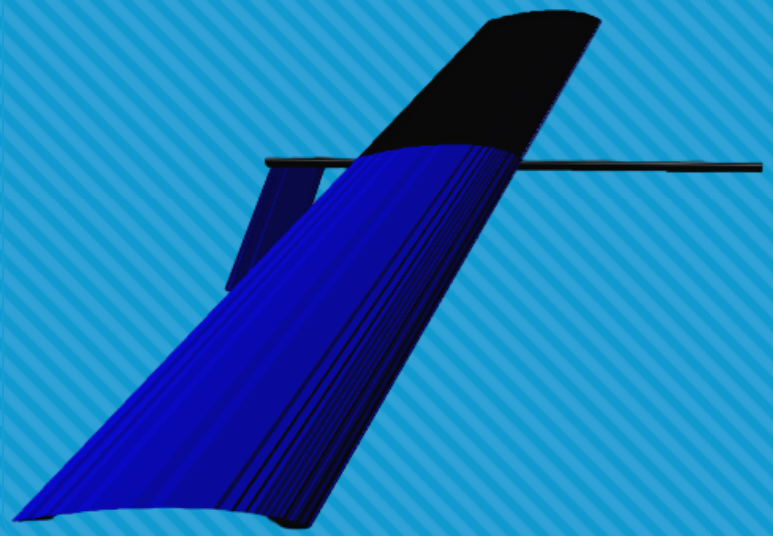
- Matlab
- Digital DatCom+

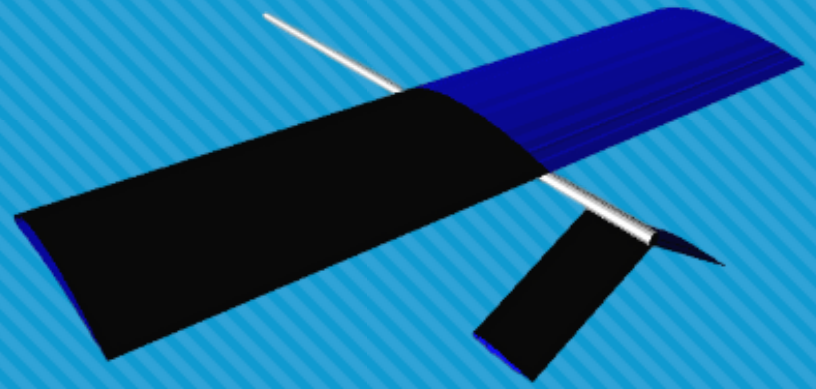


Configuration definition

- Structs definition
- Inertial properties assignment



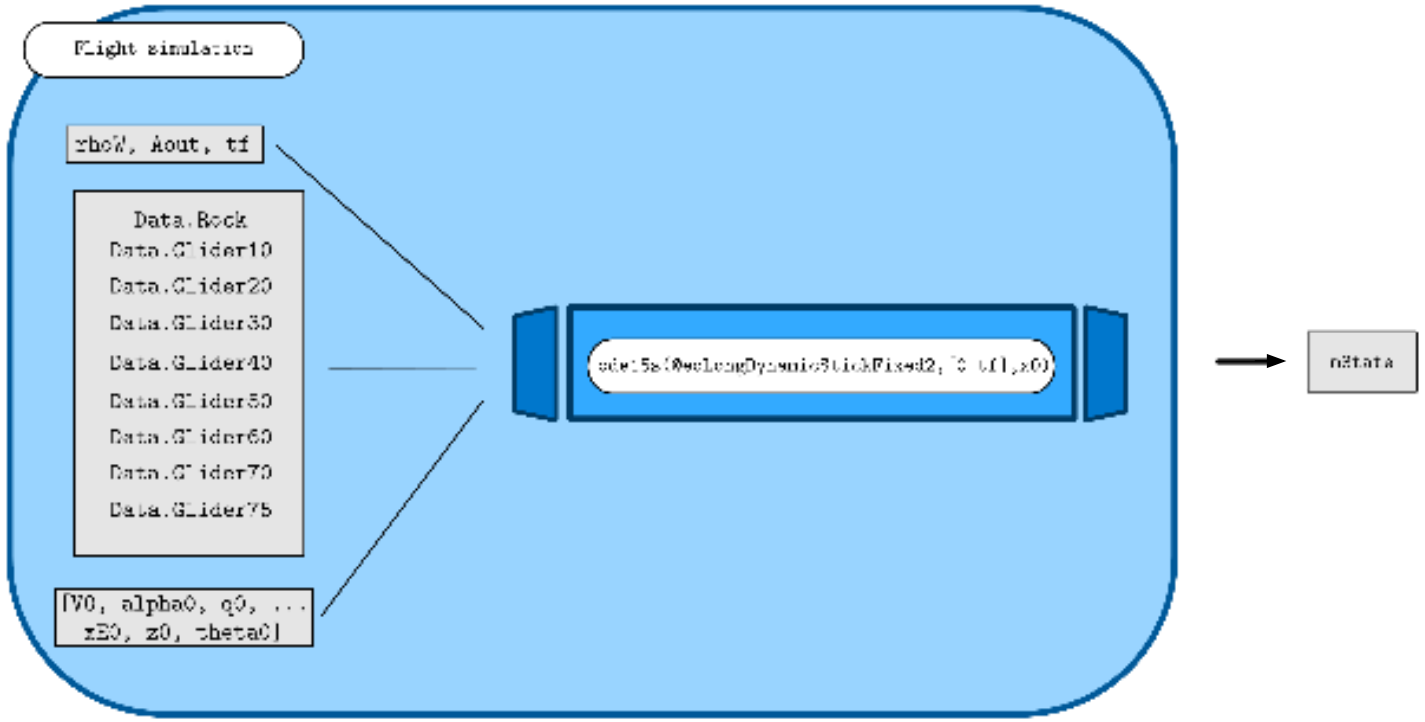




DatCom+ model

Flight simulation

- DatCom output import
- 3DoF Full System of Flight Dynamics Equation



Launch base



Requirements:

- Afford 10 bars
- Rocket support
- Rocket guide
- Fast release
- Reliability
- High safety level

Solutions:

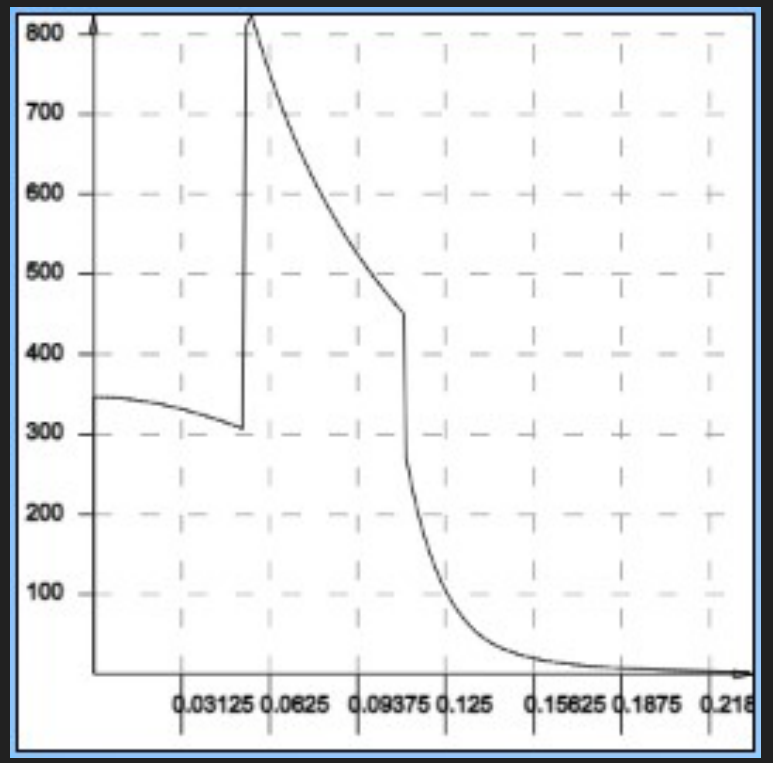
- Long launch tube
- Aluminum rod
- Glue-reinforced PVC
- Race bike valve
- Clark’s cable tie system

Propulsion system

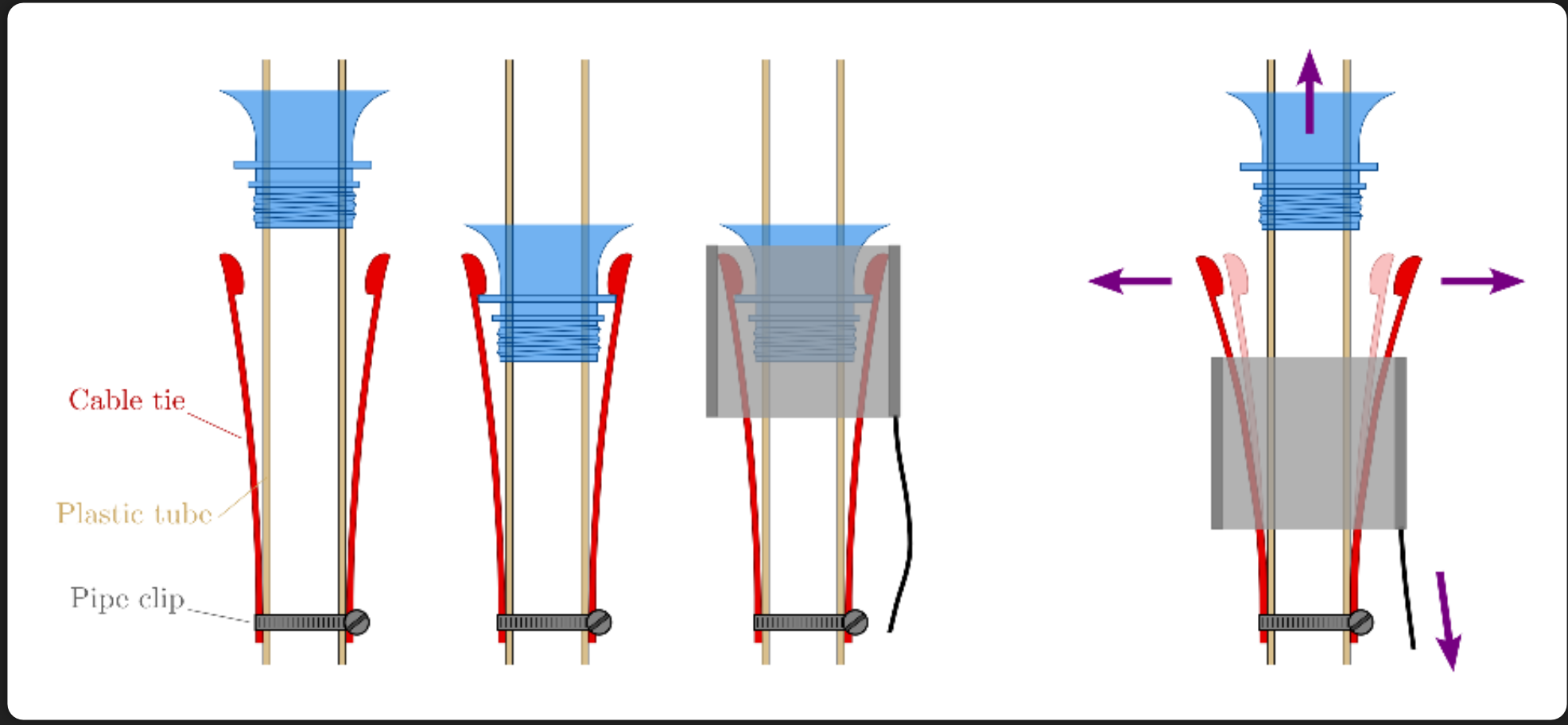
Object of study:

- Bottle shape
- Bottle dimensions
- Bottle positioning
- Launch tube lenght





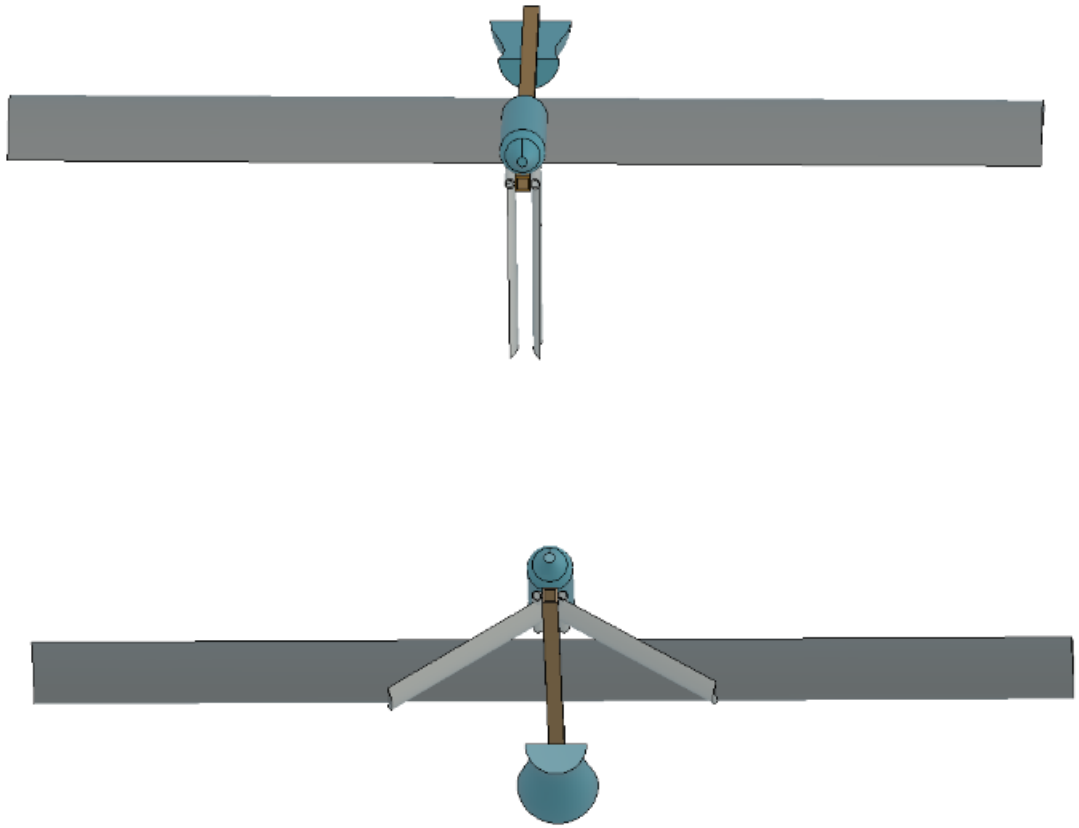
Clark's system



Control System Implementation

Flight Correction Algorithm

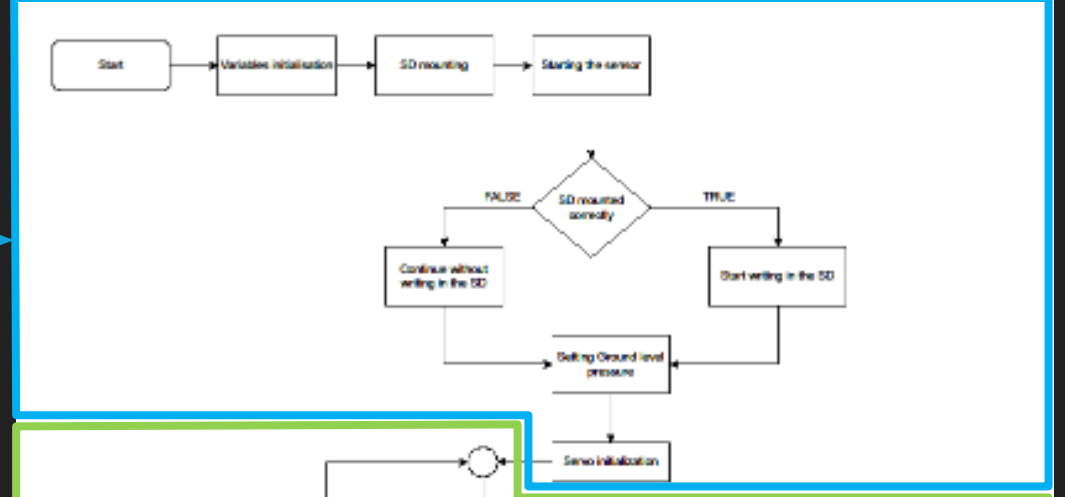
The idea is to actively stabilize the aircraft by changing the opening angle of the tailplane to dump the effects of the moment caused by the sloshing and to keep the model aircraft in a stable position.

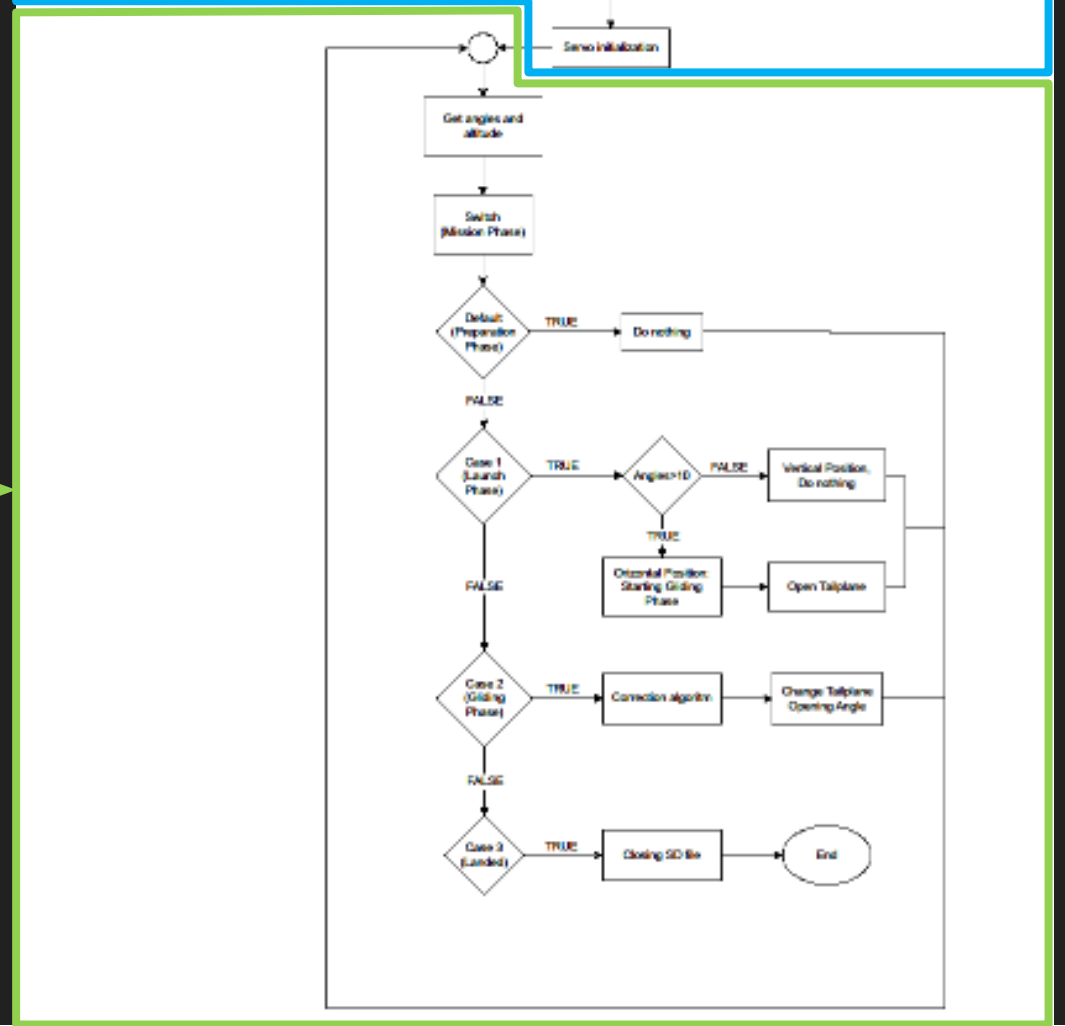


Program structure

The main itself can be divided into two parts:

- the hardware setup;
- the `while()` loop.

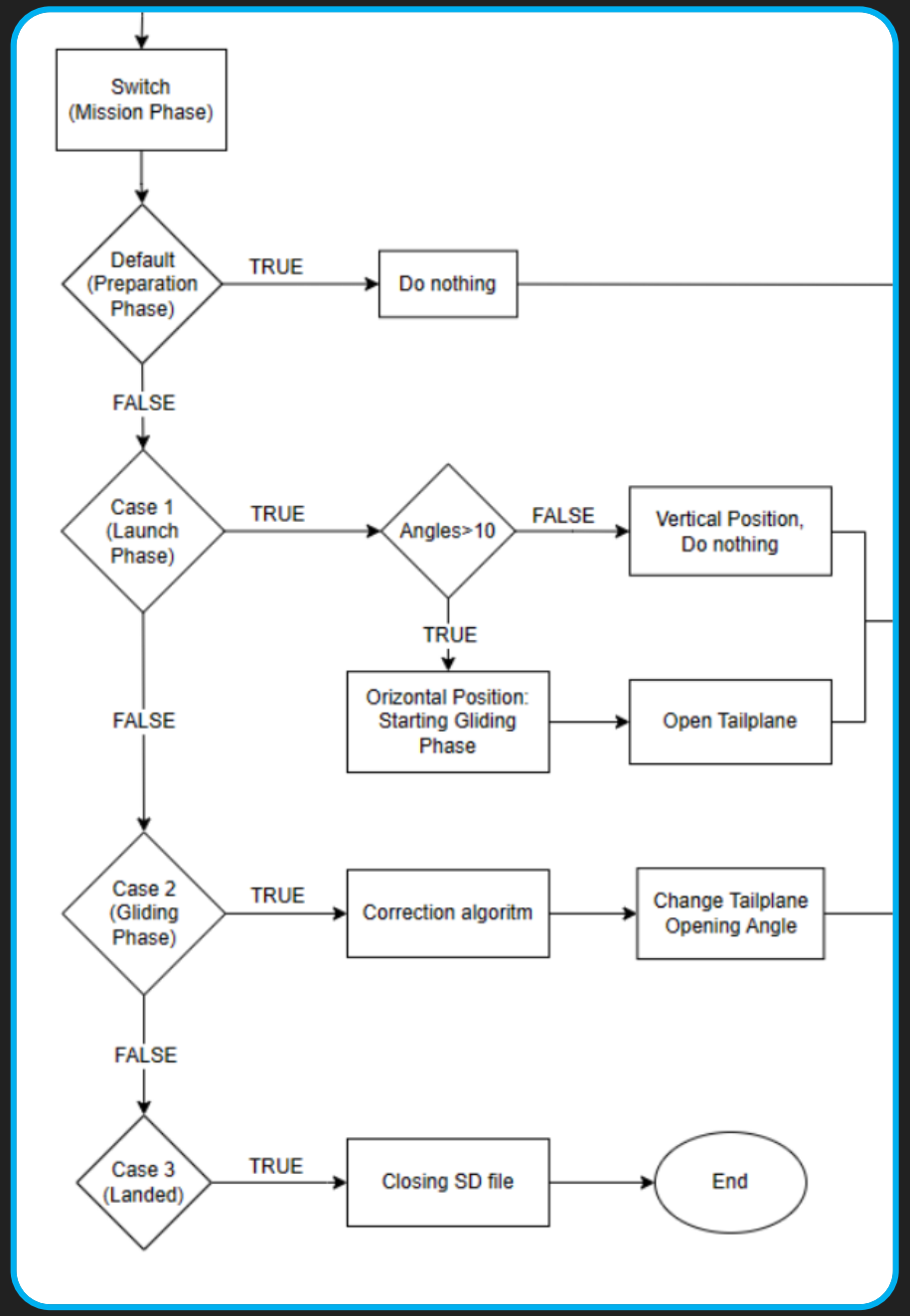




While( ):

`switch()`

- *Preparation Phase*
- *Launch Phase*
- *Glider Phase*
- *Closing Phase*



The *Glider Phase* is a flight correction algorithm that consists of a series of `if()-else()-if()` statements that determine the opening angle of the tailplane. Pitch angle values and opening angles are obtained from a series of tests.

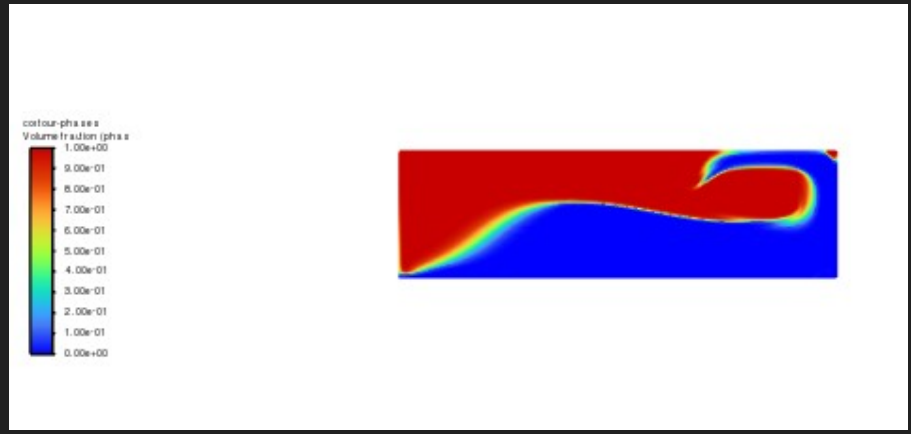
Sloshing

Sloshing

- Once the propellant water is expelled, sloshing mass may take up a large percentage of the total weight.
- The sloshing mass takes up a large fraction of the mass that significantly moves the prototype center of mass.
- The liquid exerts inertia forces on the structure mounting the tank.

The objectives are:

- Reducing the motion of water.
- Reducing the inertia forces exerted.



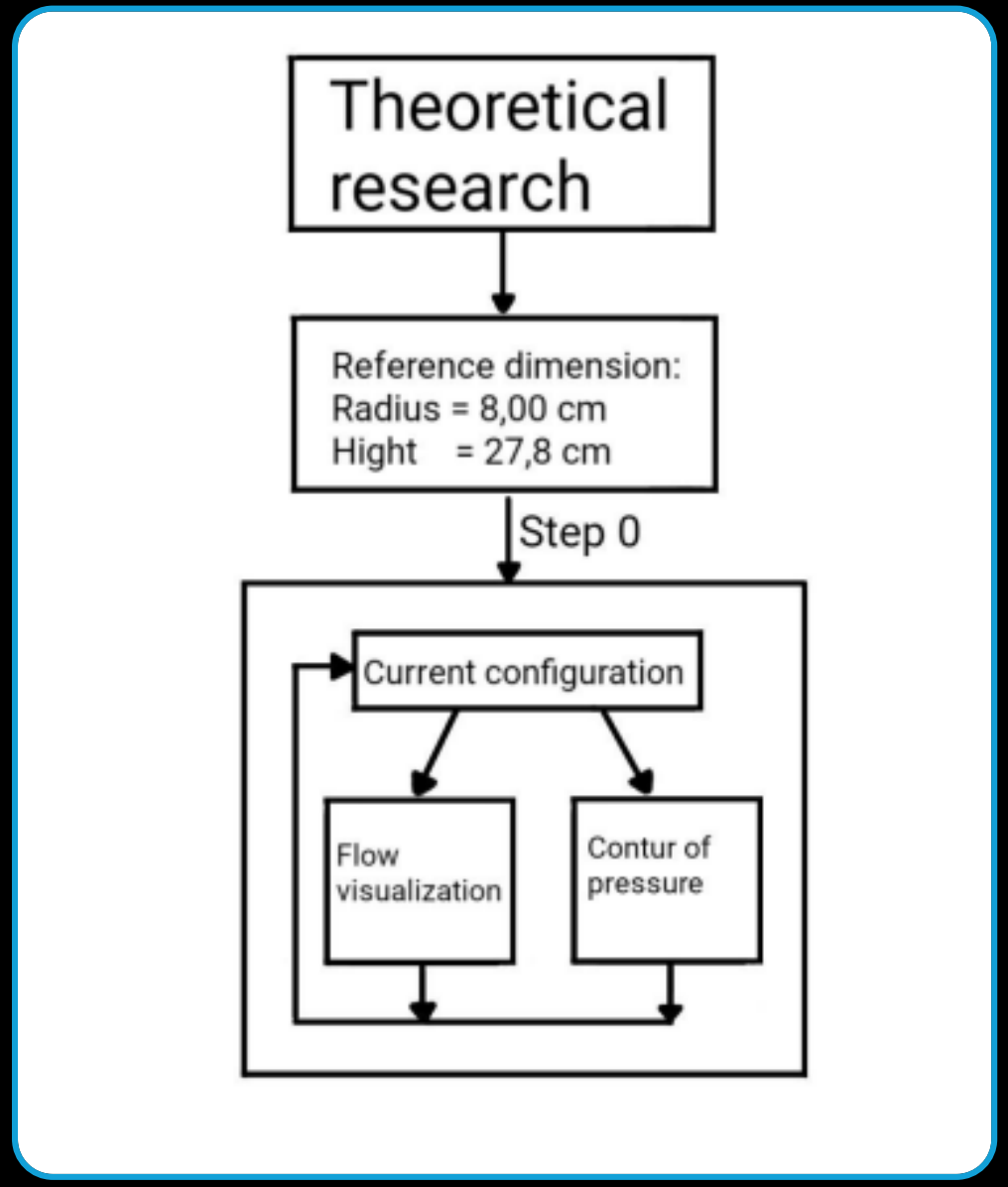
Design features to alleviate sloshing

- One 2 L bottle of propellant water (fill level at 30 % was chosen as a propulsive system: this ensures a good thrust for light structures and allows to reduce the sloshing mass, according to regulations, to 700 mL.
- Both wings and (down-loading) horizontal plane are located far behind the ideal position of the sloshing bottle in order to enhance the stability features of the model.
- Sloshing mass is an aft-located element of the configuration to maintain a stability margin against the Center of gravity shift due to sloshing.

Sloshing reduction workflow

The sloshing reduction system unfolded in three main phases:

- Conceptual design of a rocket/ glider model configuration ensuring a certain margin of static stability against sloshing movement.
- Optimization of sloshing mass carrying system by choosing a geometry.
- Sloshing mass simulation and configuration definition.



Choice of reference dimensions

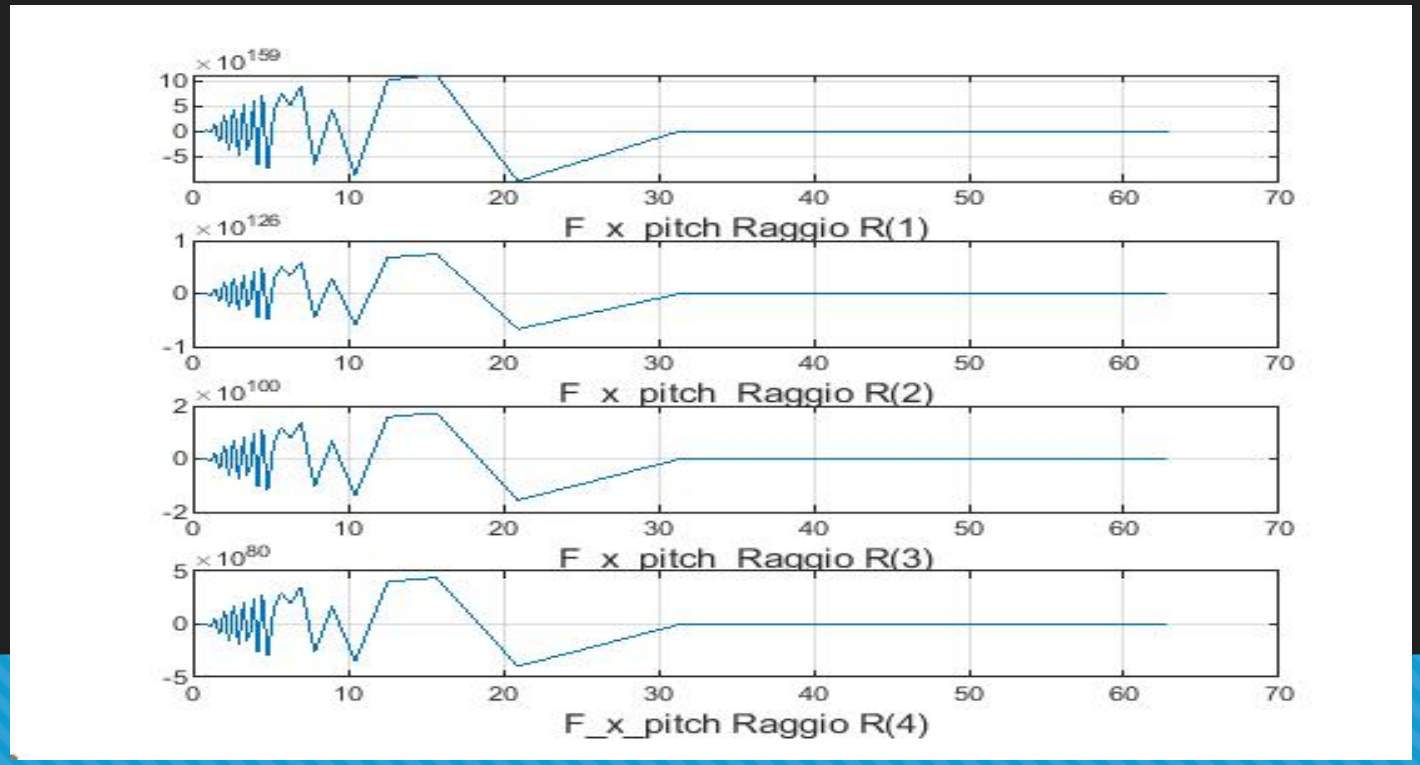
A Matlab code was implemented to extract such characteristics for the cylinder shape and to find an optimal condition, i.e., an optimal radius, considering also feasible lengths for the design.

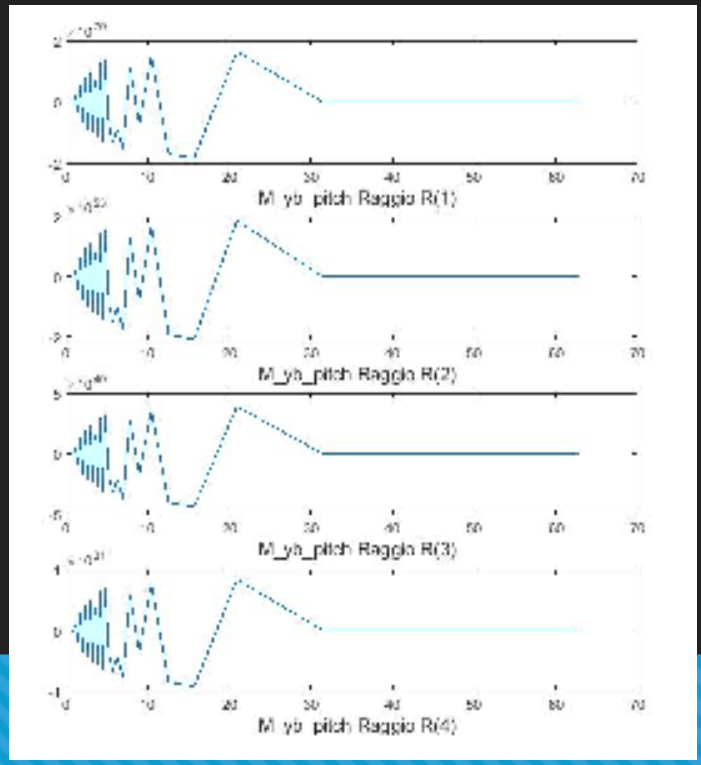
The following formula, was used, by applying Linear Sloshing Theory (the first mode ($m = 1$) was picked):

$$\omega_{mn}^2 = \frac{g\epsilon_{mn}}{R} \cdot \tanh\left(\frac{\epsilon_{mn}h}{R}\right)$$

The main considerations that may be extracted are:

- Inertia Forces decrease with increasing radius.
- Pitching moment on the walls decreases with increasing radius.

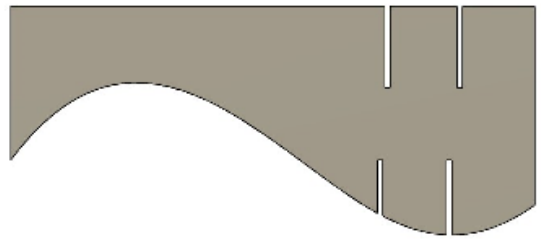




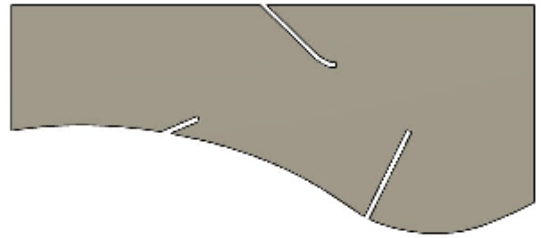
Moving tank simulation

- Configuration 1 was intended to create separated regions of "static" flow but proved to be unfavorable since it would prevent the flow from accessing outer regions at launch.

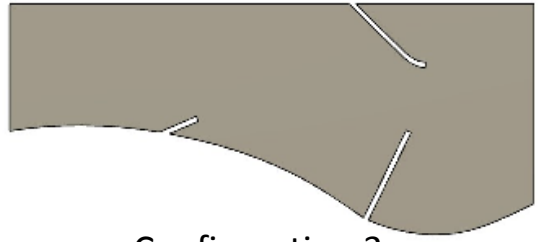
- Configuration 2 and 3 tilt the baffles to allow flow passage towards the outer basin at launch but showed some limitations since flow reaching the outer region would be stuck there. This could lead to excessive pressure on the baffles when the fluid is set in motion due to flight dynamics.



Configuration 1



Configuration 2



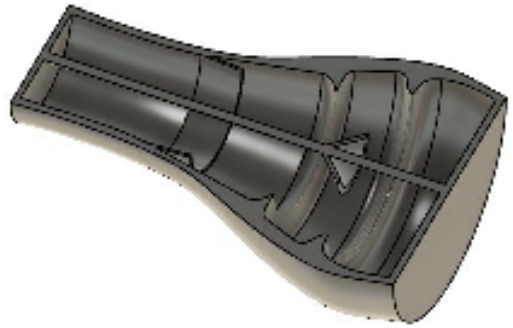
Configuration 3

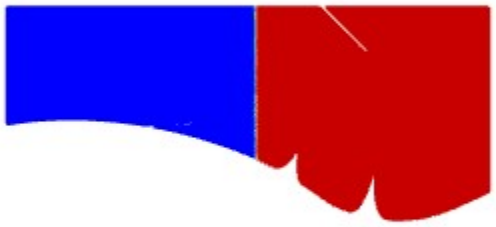
Moving tank simulation

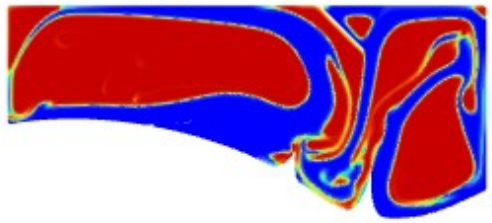


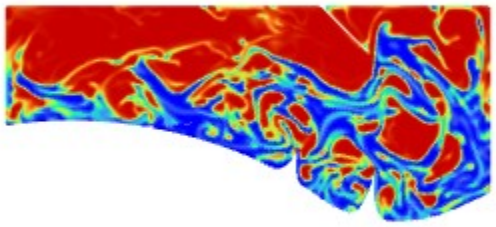
Configuration 4

- Configuration 4 was judged as a good convergence point.
- The shape is obtained through solid revolution (180°) of the configuration 4 planar shape. The total volume is computed to be 1.41 L.









Construction

Safety assessment

Here there are the identified risks in the building process.

	Manufacturing						Tests	
Activities	Wing			Fuselage		Electronics Manufacturing	Launch mechanism	
Task	Cut of Carbon fibers stick	Carbon fibers stick insertion	Hot wire wing shaping	Profiles shaping	Profiles Gluing	Soldering	Activation of launch mech.	Lift-off abort
	3	3	2	3	3	1	2	2
	2	2	2	2	2	3	2	2
	6	6	4	6	6	3	4	4

Safety assessment

Here we report the solutions employed by the team to cope with risks.

- Wing manufacturing involves risks related to the handling of sharp cutting tools in various stages. Safety gloves and safety glasses are efficient in tackling all the risks.
- While operating the hot wire, safety gloves must be worn and the operation must be carried out with careful attention and in a coordinated way.
- Gloves and Glasses are also required in the fuselage manufacturing process.
- With regards to launch mechanism, the main issues are related to being close to it at lift-off. This is tackled by using a system that can be activated from a further distance. Moreover, a security valve is employed to ensure the possibility of blowing down the tank in case of launch abort.

Materials

- EPS polystyrene (wing)
- XPS polystyrene (tailplane)

→ Since the wings aren't subjected to an excessive load, we were able to use a light material to build them without overly increasing the weight of the structure.

- Poplar plywood (fuselage)

→ This wood is light and soft, but very elastic. Its high resistance to impacts made it a perfect candidate for an important structural element as the fuselage.

- Carbon fiber (wing spars)

→ Carbon fiber has high mechanical strength and low density allowing to increase the structure of the wings without excessively increasing their weight.

Materials

- PPR (polipropilene, launch tube) → The launch tube is subjected to a pressure of 10 bar. For this reason, the material chosen is PPR, which thanks to its high resistance to deformation, is often used for pumps and valves.
- Aluminium (guide tube) → The guide tube supports the total weight of the rocket in the launch phase. Aluminum was used as material having high compressive strength.
- PET (propulsion system) → The resistance of the PET bottles is above 10 bar (the maximum condition that can be reached by regulation). To obtain the maximum yield it was decided to use this material.

Materials

- 3D printed parts → Sloshing tanks and engine mounts are 3D printed in order to reduce costs and increase accuracy.
- Electronic components → The components were choosen in order to have highly retrivable off-shelf pieces at low cost and with good performaces.

Costing

- All the materials have been chosen in order to have the highest resistance at the lowest possible price to accommodate our limited resources without loosing quality.
- Indeed, we didn’t use expensive components and they were all easily retrievable.
- The total cost of the prototype is the 406.82€, but this total includes also the cost of spare parts in case they got damaged during the manufacturing stage.

Aero-model structure	179,3
Eletronics components	131,22
Launch base	62,52
3D printed parts	33,78
Total Price	**406,82**

Design for manufacturing

The production planning started from the launch base. Once the global characteristics were defined, the construction of the prototype of the base started through a trial-and-error process.

The construction of the fuselage was achieved by designing four strips with matching ends which were cut from a 4 mm wooden sheet and then assembled to fit the strips together. The basic idea was to ensure the best anchorage.

After some research on the manufacturing of polystyrene, a hotwire device was built to cut out the wings.



Design for manufacturing



To fix the wings to the fuselage where used the carbon fibers (which gave the wings greater resistance). The fibers pass inside the fuselage and creating a joint with the polystyrene. In this way, the reliability of the structure has been increased without adding components and the structure has been lightened.

Once the main structure was obtained, the mechanical components such as engines were fixed (which were previously connected to the tailplane) and hardware.

Process of Construction

The construction process can be divided into stages that may be summoned as:

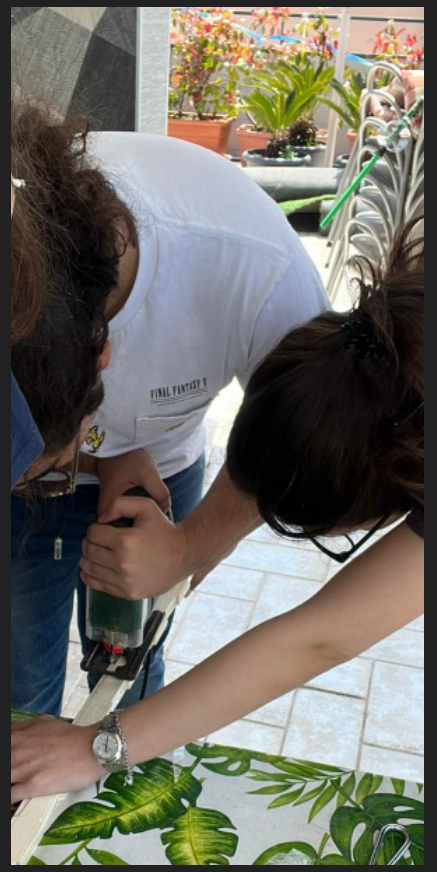
1. Launch base assembling and fuselage poplar wood cutting and assembling.
2. Wings and horizontal plane shaping via hot wire technique.
3. Fuselage-Lifting surface fixing and fuselage-motors fixing.

The launch operation consists in:

- Activate and set up the hardware.
- Pour water into the tank.
- Insert the guide tube inside the rocket and the launch tube inside the propulsive tank ensuring locking.

Manufacturing Challenges

- Building the fuselage required a lot of precision and manual sanding. → Using a grinder and a jigsaw significantly reduces the manufacturing time.
- Fitting the spars into the wing without damaging them was challenging. → Using a long enough drill can reduce the risks of damage.
- We used more glue and scotch than expected. → Using clamps may reduce the amount of wasted adhesive materials.



Lesson learnt

- **Teamwork**

  the team successfully worked in synchrony while completing parallel tasks.

- **Design**

  the challenge allowed us to test innovative concepts, other than building a functioning aircraft.

- **Flight Planning**

  the work required a careful analysis of all the flight phases and the awareness of their relation with each other.

- **Flight Test**

  last but not least, the team members handled for the first time the *Trial & Error* processes typical of the last design phases.



Thanks for your attention